

Acki Nacki Bridge

Acki Nacki Bridge — User Guide

A cross-chain bridge between **Ethereum** and **Acki Nacki**, secured end-to-end by **zero-knowledge proofs**.

Revision: 31 May 2026 · Test deployment (Sepolia)

1. What is the Acki Nacki Bridge?

The Acki Nacki Bridge lets you move value between the **Ethereum** network and the **Acki Nacki** blockchain. Unlike traditional bridges that rely on a trusted group of signers, this bridge uses **zero-knowledge (ZK) cryptographic proofs** to verify that every cross-chain event really happened. In plain terms:

You don’t have to trust an operator. The bridge proves, mathematically, that your deposit was recorded on Ethereum before any tokens are issued on Acki Nacki.

Two directions, two mechanisms

Direction	What it does	Status for users
Ethereum → Acki Nacki (Deposit)	You send ETH on Ethereum; equivalent tokens are minted to you on Acki Nacki after a ZK proof of your deposit is verified.	✅ Available
Acki Nacki → Ethereum (Withdraw / burn)	Burn tokens on Acki Nacki and release ETH on Ethereum.	⚠️ In development — see §6

⚠️ **Important:** At the time of writing, only the **Deposit** direction is available to end users. Genuine cross-chain withdrawals are still being built. Any “Withdraw” control you may see in an early build of the web app is a placeholder and will not move funds back to Ethereum.

2. Before you start

You will need:

1. **A MetaMask wallet** (browser extension or mobile). Other EIP-1193 wallets may work, but MetaMask is the supported and tested option.
2. **Test ETH on the Sepolia network.** The bridge currently runs on Ethereum's **Sepolia testnet**, not on Ethereum mainnet. Sepolia ETH has no real monetary value and is meant for testing.
3. A few minutes of patience — proof generation and verification happen automatically in the background but are not instant.

Network details (testnet)

Parameter	Value
Network name	Sepolia
Chain ID	11155111 (0xaa36a7)
Public RPC	https://rpc.sepolia.org
Block explorer	https://sepolia.etherscan.io
Bridge contract	0xDE8180911Ab2EbC9A6c1F5526bCE4c8242C061d9

💡 These values are for the current **test deployment** and may change. Always confirm the active contract address with the team before bridging anything you care about.

Getting Sepolia test ETH

You can obtain free Sepolia ETH from public faucets (search “Sepolia faucet”). You only need a small amount to try the bridge — for example, 0.01–0.1 ETH.

3. Depositing ETH (Ethereum → Acki Nacki)

This is the main user flow. Here is what happens, step by step.

What you do

1. **Open the Bridge web app** in a browser that has MetaMask installed.
2. **Connect your wallet.** Click Connect Wallet in the header and approve the MetaMask prompt.

3. **Switch to Sepolia.** If you're on another network, the app will ask MetaMask to switch to Sepolia automatically. Approve it.
4. **Enter the amount** of ETH you want to bridge in the `Deposit` form.
 - The amount must be **greater than 0** and **at most 100 ETH** per `deposit ( MAX_DEPOSIT_AMOUNT )`. A zero amount is rejected (`InvalidAmount`); anything above 100 ETH is rejected (`DepositTooLarge`).
5. **Submit and confirm** the transaction in MetaMask. This calls the bridge's `deposit()` function (it takes no arguments — the ETH you send is the deposit) and transfers your ETH to the bridge contract. Your **connected wallet address** (`msg.sender`) is recorded as the depositor and is the party credited on Acki Nacki.
6. **Wait for the transaction to be mined.** Once confirmed, the bridge emits a `Deposit` event containing your unique **Deposit ID**, your address, the amount, and a timestamp.

What happens behind the scenes

You don't have to do any of this yourself — it's fully automated — but here's what the system does so your tokens appear on Acki Nacki:

1. An **off-chain prover** reads your deposit transaction and its inclusion proof from Ethereum, then builds a **Halo2 ZK proof** (Axiom-based circuit backend) that your `Deposit` event was genuinely emitted by the bridge contract in a real Ethereum block. The proof artifacts are exported in a form the Acki Nacki verification opcode can consume directly.
2. The Acki Nacki side **verifies that proof natively** (via a dedicated TVM verification opcode) and checks the public inputs.
3. Once the proof is accepted, the corresponding **tokens are minted to you on Acki Nacki**.

The result: your ETH is locked on Ethereum, and you receive matching tokens on Acki Nacki — without trusting any human operator.

Tracking your deposit

- Use your **transaction hash** on <https://sepolia.etherscan.io> to confirm the deposit landed on Ethereum.
- Note your **Deposit ID** from the `Deposit` event — it uniquely identifies your bridging operation.

i Who receives the tokens on Acki Nacki? In the current deployment, the bridge credits the **same address that made the deposit** (`msg.sender`). A configurable Acki Nacki receiver is being prepared for a future release; until the Acki

Nacki side provides concrete receiving details, a custom receiver is tracked off-chain only and is **not** sent on-chain.

4. Checking bridge status & history

The web app surfaces live, read-only information pulled directly from the contract:

- **Total deposits** made through the bridge (`depositCount`).
- **Total value deposited** (`totalDeposited`).
- **Your transaction history**, with direct links to Etherscan.

These are informational and never require a signature.

5. Fees, limits and safety

- **Per-deposit limit:** 100 ETH maximum per `deposit()` call.
- **Gas:** You pay normal Ethereum (Sepolia) gas fees for the deposit transaction.
- **Treasury & yield:** Idle ETH held by the bridge can optionally be put to work in AAVE V3 for yield by the bridge owner. This is an operator-side feature and does not affect your deposit balance or your claim to bridged tokens.
- **Non-custodial trust model:** The bridge does not ask you to trust a signer set. Cross-chain state is advanced only when valid ZK proofs are verified on-chain.

Safety checklist

-  Confirm you are on **Sepolia** (testnet) and using the **correct contract address**.
-  Never enter your seed phrase anywhere — MetaMask never asks for it in a dApp.
-  Double-check the amount before confirming in MetaMask.
-  This is a **testnet deployment**. Do not bridge mainnet ETH or treat test tokens as having real value.

6. Withdrawals (Acki Nacki → Ethereum) — current status

True cross-chain withdrawals (burning tokens on Acki Nacki to release ETH on Ethereum) are **not yet available to users**. The legacy refund-style `withdraw` flow from earlier versions has been **retired**, and the production burn-proof flow is still under active development.

Under the hood, the Ethereum contract already carries the proof-verified machinery for the reverse channel — a permissionless `verifyBlock` entry point that advances the on-chain commitment to Acki Nacki’s state after checking ZK attestations, and a proof-gated `withdrawByProof` path — but these are protocol/operator-level and are **not** exposed as a user action yet.

What this means for you right now:

- You **can** deposit ETH and receive tokens on Acki Nacki.
- You **cannot** yet move value back from Acki Nacki to Ethereum through the bridge.
- If you see a “Withdraw” form in an early build of the app, treat it as a **non-functional placeholder**.

This guide will be updated when withdrawals go live.

7. How the bridge stays trustworthy (plain-English overview)

For the curious, here’s why the bridge is considered secure without a trusted operator:

- **Your deposit is proven, not asserted.** A zero-knowledge proof demonstrates that your deposit transaction was included in a real Ethereum block before any tokens are minted.
- **Acki Nacki’s own state is attested with proofs too.** The reverse channel checks cryptographic attestations of Acki Nacki blocks (validator signatures, block identity, and chain progression) using ZK proofs verified on Ethereum.
- **Strict on-chain rules.** Every state update enforces invariants such as monotonic block sequence numbers and chain-anchor consistency, so the bridge can’t be tricked into accepting out-of-order or forged state.

You don’t need to understand the cryptography to use the bridge — but it’s the reason the bridge can be trusted by math rather than by people.

8. Troubleshooting

Problem	Likely cause	What to do
“Please connect your wallet first”	MetaMask not connected	Click Connect Wallet and approve.
“Please switch to Sepolia network”	Wrong network selected	Approve the network switch in MetaMask, or add Sepolia manually.

Problem	Likely cause	What to do
"Invalid amount"	Empty or non-numeric amount	Enter a positive number of ETH (≤ 100).
Transaction fails / reverts	Insufficient gas, deposit over 100 ETH, or paused contract	Check your Sepolia ETH balance, lower the amount, and retry.
Deposit confirmed but no tokens on Acki Nacki yet	Proof generation/verification still in progress	Wait — minting happens after the ZK proof is verified.
MetaMask not detected	Extension missing or disabled	Install/enable MetaMask and reload the page.

9. Glossary

- **ZK proof (zero-knowledge proof):** A cryptographic proof that a statement is true without revealing extra information. Here, it proves your deposit really happened.
 - **Halo2 / Groth16:** ZK proof systems used by the bridge. Halo2 proofs are sometimes wrapped in compact Groth16 proofs to fit Ethereum's contract size limits.
 - **Deposit ID:** A unique number assigned to each deposit, emitted in the `Deposit` event.
 - **Sepolia:** An Ethereum test network used for the current deployment.
 - **TVM:** The virtual machine Acki Nacki uses to execute contracts, including native proof verification.
 - **Treasury:** ETH held by the bridge contract from user deposits.
 - **AAVE V3:** A lending protocol where idle treasury ETH may be placed to earn yield (operator-managed; does not affect your funds' claimability).
 - **Axiom (halo2-lib):** The Halo2 circuit framework the deposit prover is built on; it produces proofs the Acki Nacki verification opcode can check natively.
 - **verifyBlock:** A permissionless Ethereum-side function that advances the bridge's view of Acki Nacki's chain state after verifying ZK attestation proofs (protocol-level).
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This document describes the current test deployment of the Acki Nacki Bridge and will evolve as features such as withdrawals become available. Always verify network and contract details with the team before bridging.